MIL-C-28859A(EC)
15 October 1980
SUPERSEDING (see 6.7)
MIL-C-28859(EC)
31 December 1979

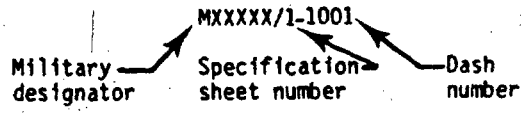
MILITARY SPECIFICATION

CONNECTOR COMPONENT PARTS, ELECTRICAL BACKPLANE, PRINTED WIRING, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the general requirements for printed wiring electrical backplane connector component parts.

1.2 Classification. The military part number shall consist of the letter M, the basic number of the specification sheet, and an assigned dash number (see 6.4) as shown in the following:



2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-C-533	Copper-beryllium Alloy Strip (Copper Alloy Numbers 170 and 172)
QQ-N-290	Nickel Plating (Electrodeposited)

MILITARY

MIL-P-13949	Plastic Sheet, Laminated, Copper Clad (For Printed Wiring)
MIL-M-24519	Molding Plastic, Polyester, Thermoplastic
MIL-C-28754	General Specification For Connectors, Electrical, Modular And Component Parts
MIL-I-43533	Epoxy Base Marking Ink
MIL-G-45204	Gold Plating (Electrodeposited)
MIL-C-45662	Calibration Systems Requirements
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assembly)
MIL-P-55110	Printed Wiring Boards
MIL-C-55330	Connector, Preparation For Delivery Of
MIL-P-55617	Plastic Sheet, Thin Laminate, Copper-Clad (For Printed Wiring, Primarily For Multilayer)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5935

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure And Tables For Inspection By Attributes
MIL-STD-202	Test Methods For Electronic And Electrical Component Parts
MIL-STD-1285	Marking Of Electrical And Electronic Parts
MIL-STD-1344	Test Methods For Electrical Connectors

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE/AMERICAN SOCIETY FOR TESTING MATERIALS

ANSI/ASTM B 122-79	Standard Specification for Copper-Nickel-Tin-Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver), And Copper-Nickel Alloy Plate, Sheet,Strip And Rolled Bar

(Application for copies should be addressed to the American National Standards Institute, 10 East 40th Street, New York, NY 10016.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.4).

3.2 Qualification. Connector component parts furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of bids (see 6.5).

3.3 Materials. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the connector assembly to conform to the performance requirements. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the furnished product.

3.3.1 Compliant component material. The compliant components shall be made from the following material:

a. Compliant pin: Beryllium copper sheet in accordance with QQ-C-533, alloy 172, with a minimum hardness of 50 on the Rockwell 30N scale. The compliant pin shall have a minimum Knoop microhardness of 310.

b. Feed-to and Feed-through: Copper alloy sheet, UNS number C72500, temper H08, in accordance with ANSI/ASTM B 122-79.

3.3.1.1 Compliant component finish. The compliant components shall be finished with gold plate in accordance with MIL-G-45204, Type II, Grade C, Class 1, to a thickness of 50 microinches minimum over nickel plating in accordance with QQ-N-290, Class 1, to a thickness of 50 microinches minimum. Exposed base material is permitted in noncritical break off or plating carrier areas of the connector parts providing all performance requirements are maintained with no degradation of critical areas (see 3.1).

3.3.2 Housing material. The housing material shall be polyester thermoplastic in accordance with MIL-M-24519, Type GPI-30F, color black.

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3.4 Design requirements.

3.4.1 Compliant components. The compliant components shall be designed in accordance with the applicable specification sheet. The design of the compliant feature shall be the responsibility of the individual manufacturer. The compliant pin component shall be capable of being individually removed without removing the housing.

3.4.2 Housing. The housing shall be designed in accordance with the applicable specification sheet. The housing shall be designed to be removed and replaced without removing the compliant pin from the printed wiring backplane.

3.5 Performance requirements.

3.5.1 Housing retention. The housing retention force shall be as specified (see 3.1).

3.5.2 Compliant component retention.

3.5.2.1 Initial. After initial installation of the compliant component into the printed wiring backplane the maximum pushout force allowed shall be 45 pounds (200.17 Newton), when tested in accordance with 4.7.3.2.

3.5.2.2 Conditioned. After conditioning in accordance with 4.7.3.2.1, the pushout force shall be a minimum of 7.5 pounds (33.36 Newton).

3.5.3 Plated-through hole integrity. When microsectioned in accordance with 4.7.3.6, plated-through holes containing compliant components shall conform to the requirements specified in 3.5.3.1 through 3.5.3.3.

3.5.3.1 Hole deformation radius. The average plated-through hole deformation radius shall be no greater than 0.0015 inches (0.0038 centimeters (cm)), when measured from the drilled hole, based on a 10-hole microsection sample. The absolute maximum deformation radius shall be 0.002 inches (0.0051 cm) (see FIGURE 1).

3.5.3.2 Hole wall damage. When measured in accordance with 4.7.3.6, the minimum copper thickness remaining between the compliant component and the printed wiring laminate, averaged over a 10-hole microsection sample, shall not be less than 0.0003 inch (.0008 cm). In addition, there shall be no copper cracks or other interplane separations from the hole wall barrel or separations between the printed wiring board laminate and the plated copper barrel.

3.5.3.3 Vertical plane. The sample shall be viewed to ensure that no copper cracks, separations between conductor interfaces, or laminate-to-copper separations have occurred.

3.5.4 Electrical resistance.

3.5.4.1 Contact resistance. When the contacts are tested as specified in 4.7.2.1 and FIGURE 2, the voltage drop shall not exceed 20 millivolts (mV).

3.5.4.2 Contact to printed wiring backplane resistance. When the contacts are tested as specified in 4.7.2.2 and the applicable figure, the voltage drop shall not exceed 6mV for double sided boards (FIGURE 3), and 20 mV for multilayer boards (FIGURE 4).

3.5.4.3 Insulation resistance. When connectors are tested as specified in 4.7.2.3, the insulation resistance shall be as specified in TABLE I.

TABLE I. Insulation resistance (megohms).

Before humidity	Measured within 1 hour after removal from chamber	After 2 hours drying
10,000	10	500

3.5.5 Engaging and separating force. When tested as specified in 4.7.3.1, the force to engage shall be 10 ounces (2.78 Newton) maximum, and the force to separate shall be 2 ounces (0.56 Newton) minimum. The test shall be performed with the housing assembled.

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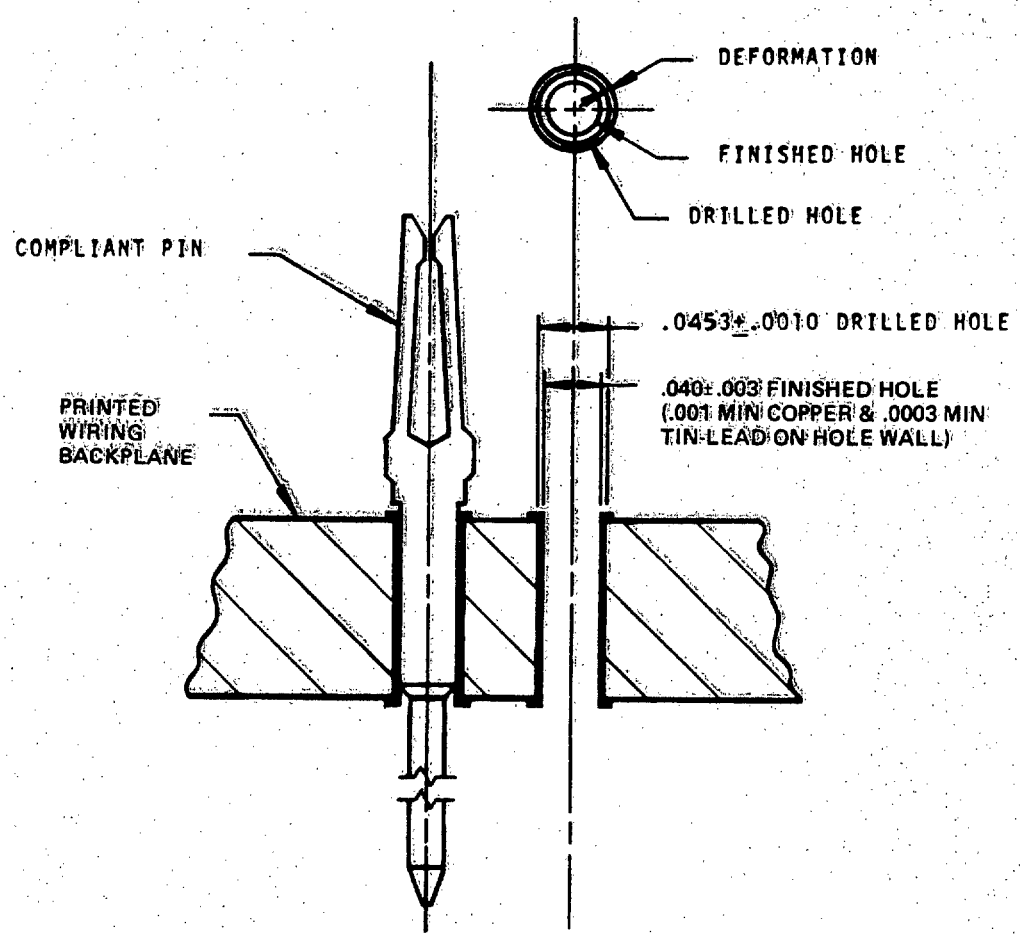


FIGURE 1. Plated-through hole deformation radius.

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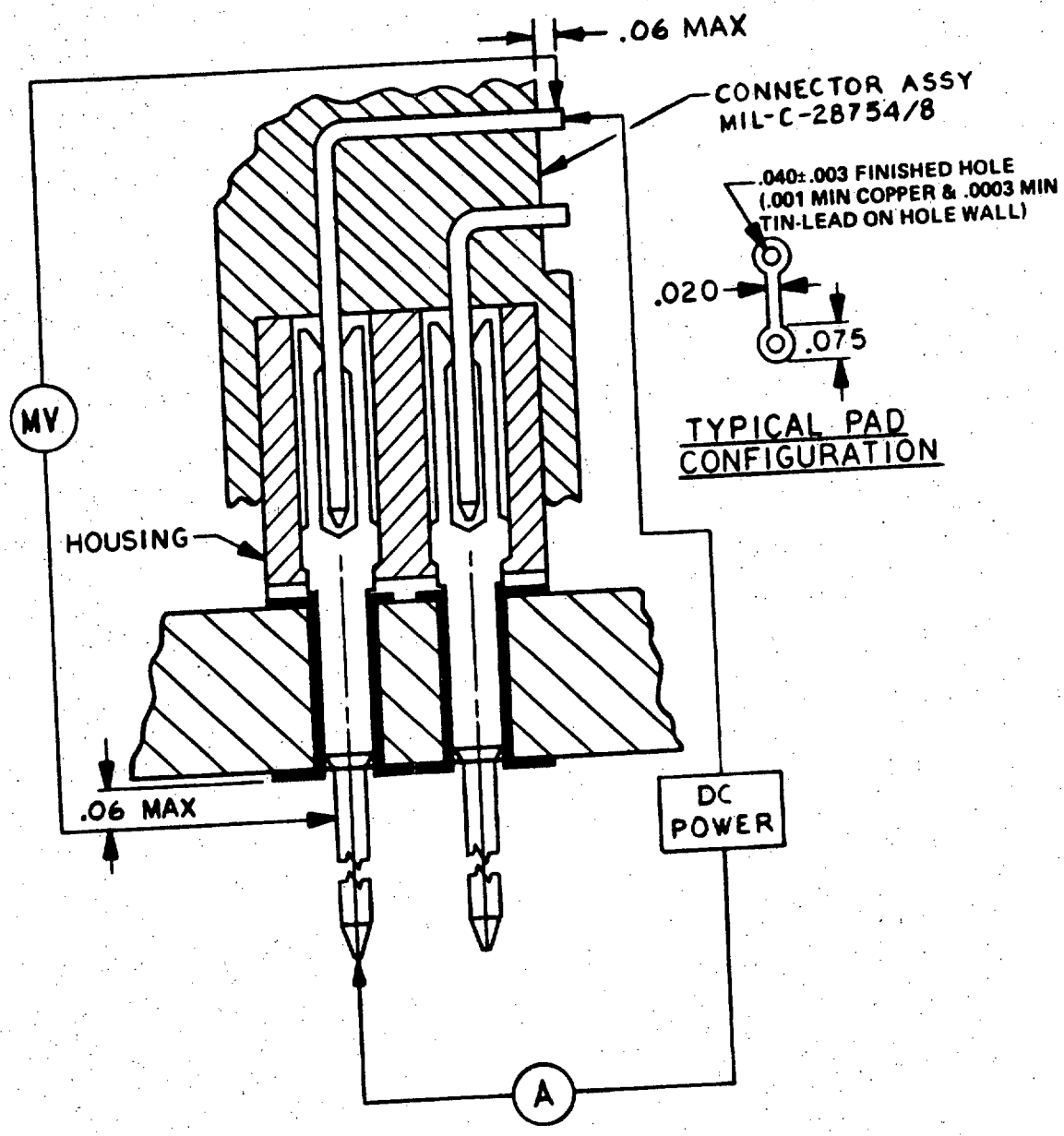


FIGURE 2. Test setup for measuring mated contact resistance.

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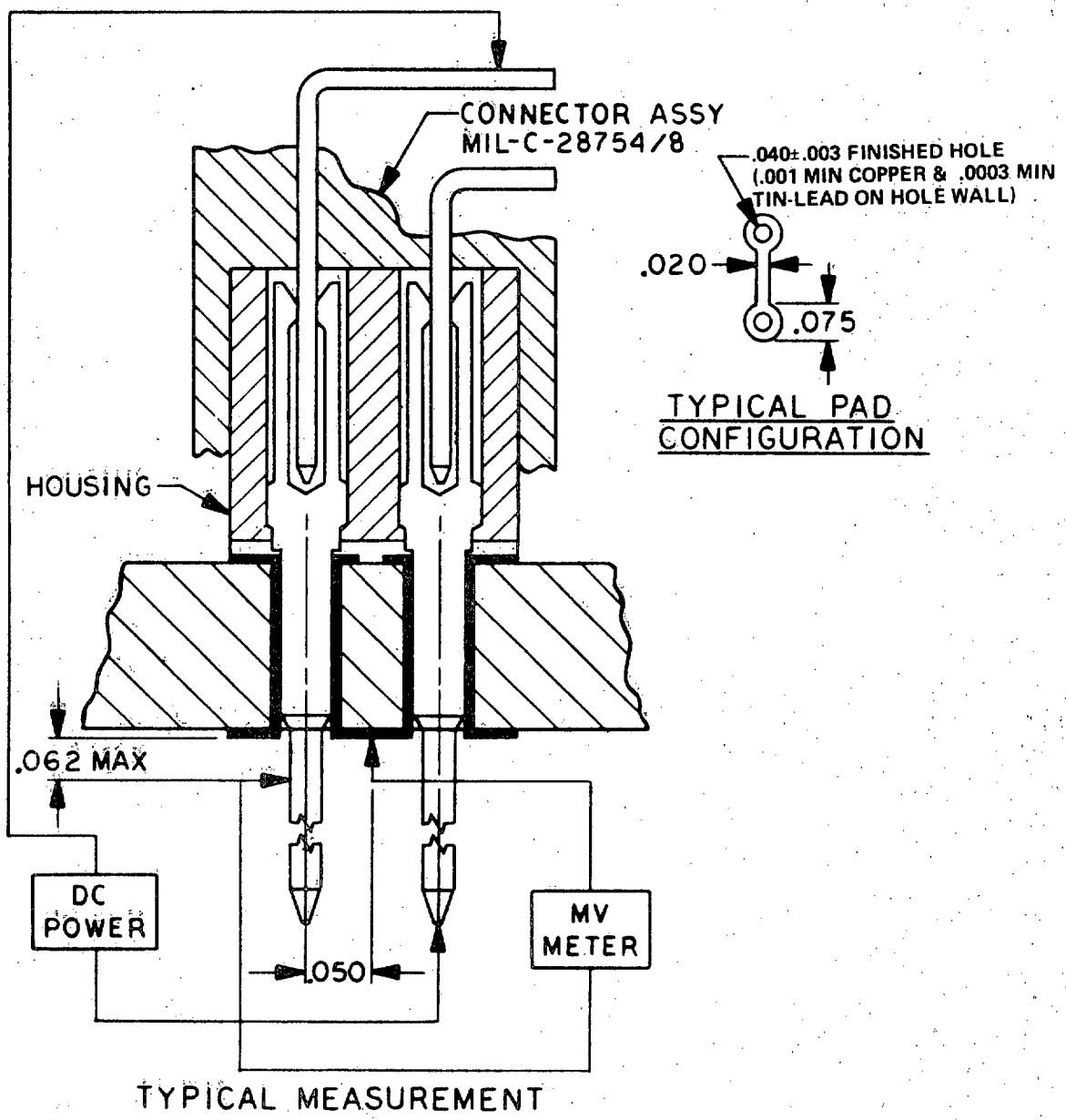


FIGURE 3. Contact to plated-through hole resistance (double sided boards).

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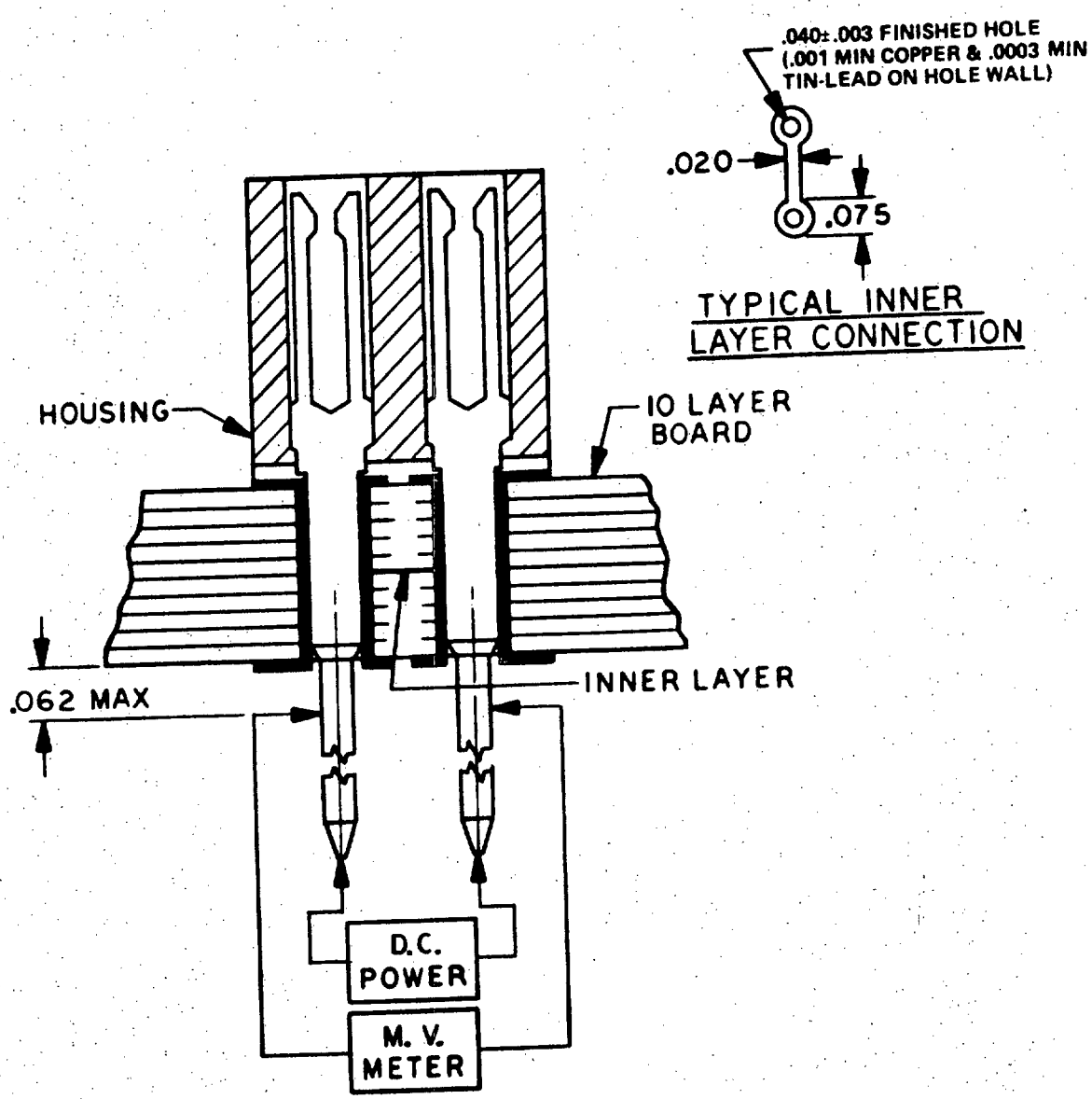


FIGURE 4. Contact to inner layer resistance (multilayer boards).

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3.5.6 Compliant component torque. When tested as specified in 4.7.3.3 the component shall withstand a minimum torque of 3 ounce-inches (0.02119 Newton-meter). Following removal of the applied torque no displacement or deformation of the compliant component shall be visible.

3.5.7 Thermal shock. When the connector assembly is tested as specified in 4.7.4.1 the connector assembly shall show no evidence of cracking, fracturing, or other damage detrimental to the operation of the connector assembly.

3.5.8 Vibration. When connector component parts assembled onto the qualification test sample (see FIGURE 5) and mated with connector assemblies qualified to MIL-C-28754 and with an appropriate supporting structure, are tested as specified in 4.7.4.4, there shall be no cracking or breaking nor shall there be any loosening of parts or other visible damage. There shall be no loss of electrical continuity of any of the contact circuits of more than one microsecond (μs) during test. Connectors shall conform to the requirements of 3.5.4 and 3.5.5.

3.5.9 Shock (specified pulse). When connector component parts assembled onto the qualification test sample (see FIGURE 5) and mated with connector assemblies qualified to MIL-C-28754 and with an appropriate supporting structure, are tested as specified in 4.7.4.5, there shall be no breakage or loosening of contacts, breaking or cracking of housings, nor other visible or dimensional damage which could cause electrical or mechanical breakdown. There shall be no loss of continuity of any of the contact circuits of more than 1 μs during test.

3.5.10 Salt spray (corrosion). When connector assemblies are tested as specified in 4.7.4.6, there shall be no visual signs of corrosion on the connector assembly that may degrade the mechanical or electrical performance.

3.5.11 Durability. When connector component parts assembled onto the qualification test sample (see FIGURE 5) and mated with connector assemblies qualified to MIL-C-28754 and with an appropriate supporting structure, are tested as specified in 4.7.3.5, there shall be no evidence of damage. In addition, after the test, the connector assembly shall conform to the requirements of 3.5.4.1 and 3.5.5.

3.5.12 Moisture resistance. When tested as specified in 4.7.4.2, the connector assembly shall conform to the requirements of 3.5.2, 3.5.4.2, 3.5.4.3, and 3.5.13.

3.5.13 Dielectric withstanding voltage. When the connector assembly is tested as specified in 4.7.2.4, there shall be no evidence of arcing, breakdown, or damage.

3.5.14 Temperature-altitude. When tested as specified in 4.7.4.3, there shall be no evidence of cracks, burns, or other visible or dimensional damage which could cause electrical or mechanical breakdown of the connector assembly. In addition, at the completion of the temperature-altitude test the connector assembly shall conform to the requirements of 3.5.4.3 and TABLE I (before humidity) and 3.5.13.

3.5.15 Life. When tested as specified in 4.7.4.7 the connector component parts shall exhibit no evidence of cracks, burns, or other visual or mechanical damage.

3.5.16 Gas-tight-joint. When tested as specified in 4.7.4.8 the connector component parts shall conform to the requirement of 3.5.4.2.

3.6 Marking. Unless otherwise specified (see 3.1) connector component parts shall be permanently and legibly marked with the military part number (see 1.2), date code, and manufacturer's source code in accordance with MIL-STD-1285. Other markings which in any way interfere with, obscure, or confuse those specified herein are prohibited. All markings shall be made using epoxy base ink in accordance with MIL-I-43553 and shall remain legible after all tests. For components too small to mark, marking shall be on the packaging material (see 5.1).

3.7 Workmanship. The workmanship on connector component parts supplied under this specification shall be suitable to assure conformance with the requirements contained herein. The compliant components shall be free of burrs or sharp corners which would otherwise damage the plating of mating connectors. Housing shall be free of cracks, voids, irregularities, sharp edges, or flash.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3)
b. Qualification inspection (see 4.5)
c. Quality conformance inspection (see 4.6)

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in TABLE II, used in fabrication, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE II. Materials inspection.

Material	Requirement paragraph	Applicable specification
Copper alloy	3.3.1	QQ-C-533 and ANSI/ ASTM B 122-79
Plastic	3.3.2	MIL-M-24519
Nickel-plating	3.3.1.1	QQ-N-290
Gold-plating	3.3.1.1	MIL-G-45204
Marking ink	3.6	MIL-I-43553

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the GENERAL REQUIREMENTS of MIL-STD-202, and MIL-STD-1344.

4.5 Qualification inspection. Qualification inspection shall be performed annually at a laboratory acceptable to the Government (see 6.5) on sample units produced with equipment and procedures normally used in production.

4.5.1 Sample.

4.5.1.1 Sample size. The qualification sample shall consist of three fully pinned printed-wiring electrical backplane assemblies in accordance with FIGURE 5 plus individual piece parts (that is, compliant components and housings), that are representative of production hardware. A minimum of nine double row hole patterns shall contain the compliant pin components and housings. The last double row pattern shall contain one row of feed-to and one row of feed-through compliant components, if these are to be qualified. The last double row shall contain compliant pin components and housings if the feed-to and feed-through compliant components are not to be qualified. The qualifying activity will supply the unpinned printed wiring board conforming to MIL-P-55110 and FIGURE 5. The wrappost length of the compliant components installed in the backplane assembly shall be 0.50 inch (.127 cm). Ten of each compliant component to be qualified and 50 housings shall be supplied in addition to the fully pinned backplane assemblies.

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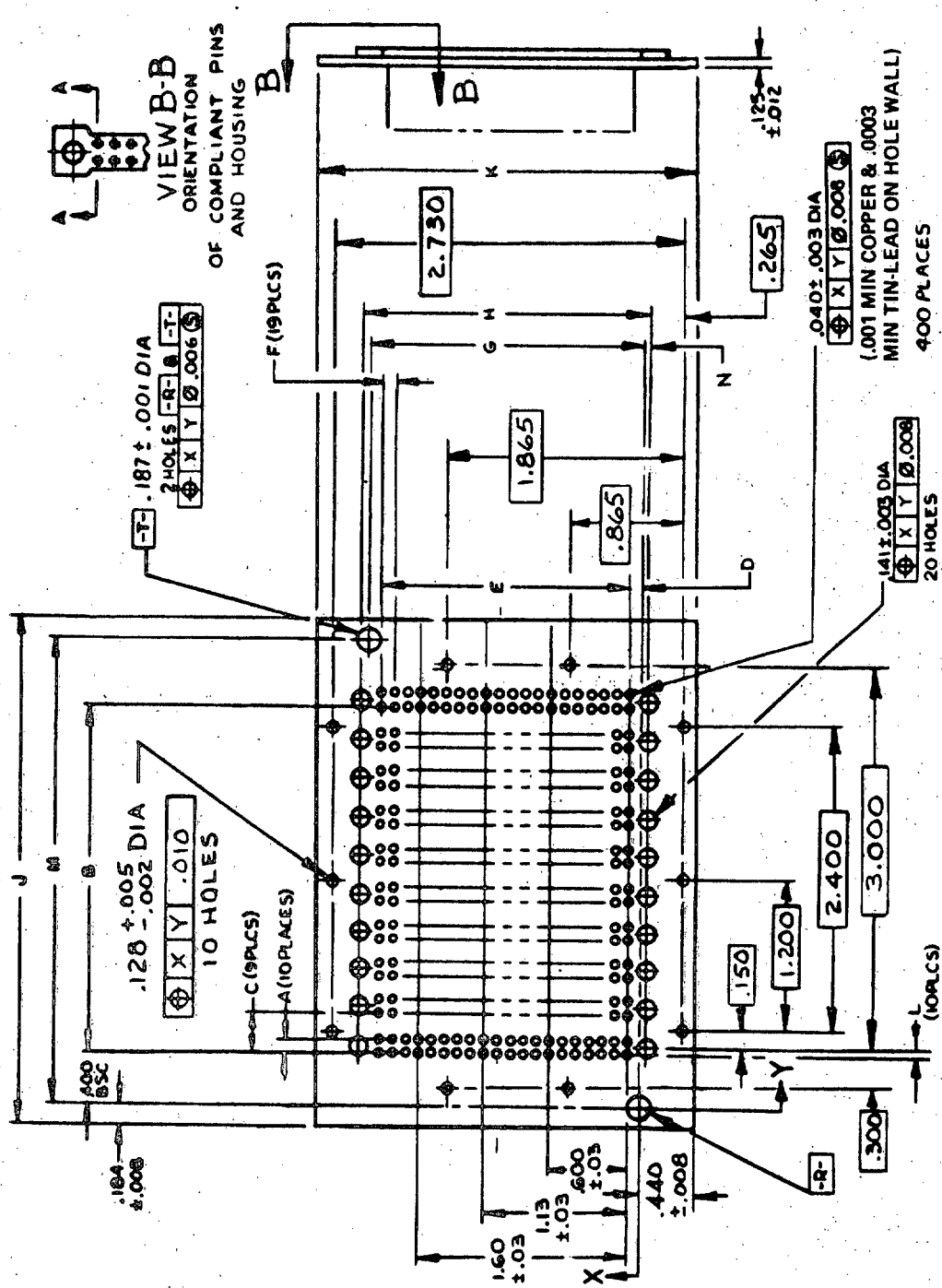


FIGURE 5. Qualification test sample configuration.

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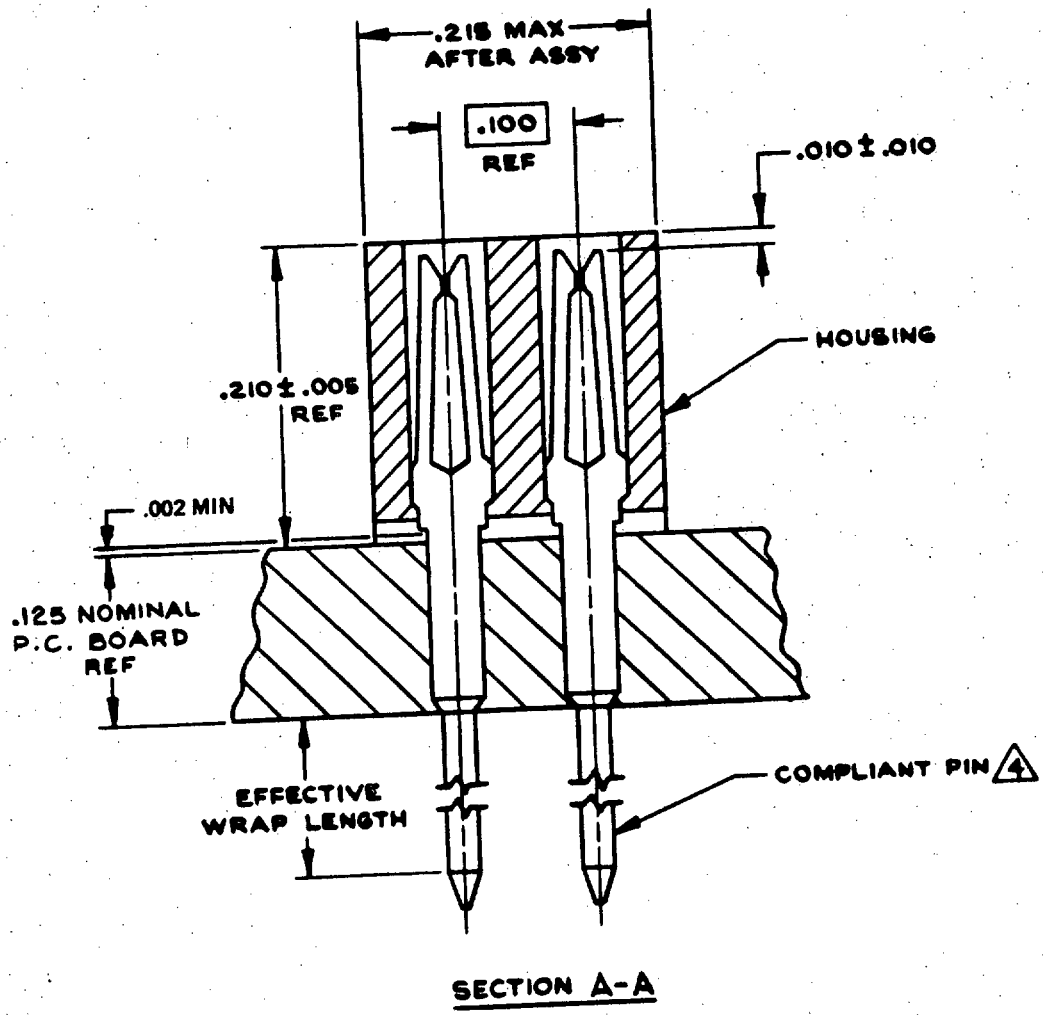


FIGURE 5. Qualification test sample configuration. (Continued)

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GRID SPACING	A BSC	B BSC	C BSC	D BSC	E BSC	F BSC	G BSC	H BSC	J ±.01	K ±.01	L BSC	M BSC	N BSC
.100	.100	2.700	.300	.100	1.900	.100	2.100	2.200	3.97	2.98	.050	3.600	.050

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.0005	.01	.0725	1.84	.1875	4.76	2.100	53.34	3.375	85.72
.0015	.04	.075	1.90	.300	7.62	2.200	55.88	3.6000	91.44
.003	.08	.080	2.03	.375	9.52	2.375	60.32	3.888	100.79
.005	.13	.100	2.54	.400	10.16	2.625	66.68	4.050	102.87
.008	.20	.125	3.18	.450	11.43	2.700	68.58	4.300	109.22
.020	.51	.141	3.58	.600	15.24	2.725	69.21	4.668	118.57
.03	.76	.150	3.81	1.13	28.70	2.850	72.39	5.000	127.00
.050	1.27	.173	4.39	1.60	40.64	3.150	80.01	5.368	136.35
.0325	1.59	.184	4.67	1.900	48.26	3.250	82.55		

NOTES:

1. Test sample shall be conformal coated using MIL-I-46058, type UR coating material prior to contact installation. Coating thickness shall be .001 inch - .003 inch when measured on a flat unencumbered surface.
2. Copper clad laminate for the test samples shall conform to MIL-P-13949, type GF for double-sided boards and to MIL-P-55617, type GF for multilayer boards. Inner layer copper thickness for multilayer boards shall be 2.0 ounces per square foot.
3. Double-sided and multilayer printed wiring boards will be provided by the qualifying activity upon request from the pin/housing manufacturer. These boards shall then be pinned and housings shall be assembled by the compliant pin manufacturer. Test samples shall then be forwarded to the qualifying activity for qualification testing. Assembly of pins and housings shall conform to FIGURE 1. Printed wiring boards shall conform to the requirements of MIL-P-55110.

△4. Test samples shall incorporate compliant pins conforming to MIL-C-28859/1-05 (0.500 inch wrappost length).

FIGURE 5. Qualification test sample configuration. (Continued)

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4.5.2 Inspection routine. Sample units shall be subjected to the qualification inspection specified in TABLE III, in the order shown.

4.5.3 Failures. Failures in excess of those allowed in TABLE III shall result in refusal to grant qualification approval.

TABLE III. Qualification inspection.

Examination or test	Requirement paragraph	Test paragraph	Number to be inspected	Number of defects allowed
Subgroup I			1	0
Visual examination	3.1, 3.3, 3.4, 3.6, 3.7	4.7.1		
Engaging and separating forces	3.5.5	4.7.3.1		
Life	3.5.15	4.7.4.7		
Thermal shock	3.5.7	4.7.4.1		
Moisture resistance	3.5.12	4.7.4.2		
Housing retention	3.5.1	4.7.3.4		
Subgroup II			1	0
Visual examination	3.1, 3.3, 3.4, 3.6	4.7.1		
Contact retention (initial)	3.5.2.1	4.7.3.2		
Hole conditioning	3.5.2.2	4.7.3.2		
Life	3.5.15	4.7.4.7		
Thermal shock	3.5.7	4.7.4.1		
Contact retention (final)	3.5.2.2	4.7.3.2		
Component torque	3.5.6	4.7.3.3		
PTH-contact resistance	3.5.4.1	4.7.2.1		
PTH-integrity (microsection)	3.5.3	4.7.3.6		
Gas-tight-joint	3.5.16	4.7.4.8		
PTH-contact resistance	3.5.4.1	4.7.2.1		
Subgroup III			1	0
Visual examination	3.1, 3.3, 3.4, 3.6	4.7.1		
Dielectric withstanding voltage	3.5.13	4.7.2.4		
Mated contact resistance	3.5.4.1	4.7.2.1		
Durability	3.5.11	4.7.3.5		
Temp/altitude	3.5.14	4.7.4.3		
Shock	3.5.9	4.7.4.5		
Vibration	3.5.8	4.7.4.4		
Moisture resistance	3.5.12	4.7.4.2		
Mated contact resistance	3.5.4.1	4.7.2.1		
Salt spray	3.5.10	4.7.4.6		

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TABLE III. Qualification inspection. (Continued)

Examination or test	Requirement paragraph	Test paragraph	Number to be inspected	Number of defects allowed
Subgroup IV			20 pins 10 housings	0
Visual examination	3.1, 3.3, 3.4, 3.6	4.7.1		
Dimensional	3.1	4.7.1		
Material	3.3.1 3.3.2			
Pin hardness	3.3.1			
Housing, retention (gage)	3.5.1	4.7.3.4		

4.5.4 Retention of qualification. To retain qualification, the manufacturer, as directed by the qualifying activity, shall forward every 12 months the qualification sample as specified in 4.5.1.1 to the qualifying activity.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Groups A and B inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all components of the same type, produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations specified in TABLE IV.

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection Level II. The acceptable quality level (AQL) shall be as specified in TABLE IV.

TABLE IV. Group A inspection.

Examination	Requirement paragraph	Test paragraph	AQL (percent defective)
Visual and mechanical Marking	3.1, 3.3.1, 3.7 3.6	4.7.1 4.6.2	2.5

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be kept separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection. Group B inspection shall consist of the examinations and tests specified in TABLE V, in the order shown, and shall be made on sample units which have been subjected to and have passed the Group A inspection. In lieu of inserting the compliant pin into a printed wiring backplane to perform the requirements of the Group B inspections, the manufacturer may elect to build a permanent test fixture which would capture the compliant feature of the compliant pin and simulate the resting surface for the housing.

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4.6.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection Level S-4, based on contacts and housings. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The AQL shall be as indicated in TABLE V.

TABLE V. Group B inspection.

Examination	Requirement paragraph	Test paragraph	AQL (percent defective)
Engaging and separating forces Housing retention	3.5.5 3.5.i	4.7.3.1 4.7.3.4	0.4

4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be kept separate from new lots, and shall clearly be identified as reinspected lots.

4.6.1.3.3 Disposition of sample units. Sample units which have been subjected to Group B inspection shall not be delivered on the contract.

4.6.2 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. Connector component parts (housings and compliant components) shall be examined to determine that the material, design, construction, physical dimensions, markings, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.6, 3.7).

4.7.2 Electrical testing.

4.7.2.1 Contact resistance (see 3.5.4.1). Compliant pins shall be wired in accordance with Method 3004 of MIL-STD-1344. The following details and exception shall apply:

a. Method of connection: See FIGURE 2
b. Test current: Three amperes
c. Twenty contacts per sample shall be tested.

4.7.2.2 Plated-through hole to contact resistance (see 3.5.4.2). Compliant pins shall be tested in accordance with the test set-up depicted in FIGURE 3. The following details shall apply:

a. Method of connection : See FIGURES 3 and 4
b. Test current: Three amperes
c. Twenty contacts per sample shall be tested.

4.7.2.3 Insulation resistance (see 3.5.4.3). Compliant pins and housings shall be tested in accordance with Method 3003 of MIL-STD-1344. The following details and exception shall apply:

a. Test voltage: 100 volts direct current (VDC) ± 5 percent
b. Test points: Between adjacent contacts on double-sided boards

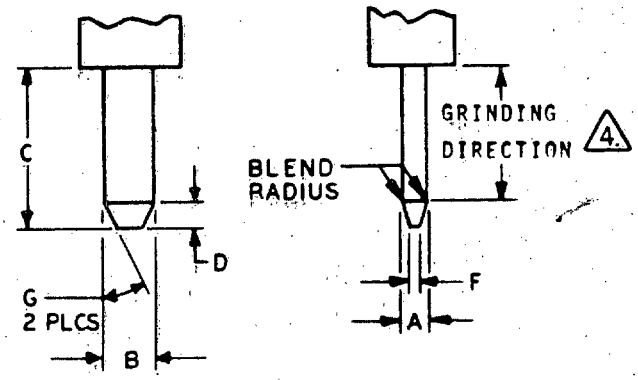
4.7.2.4 Dielectric withstanding (see 3.5.13). Compliant pins and housings shall be tested in accordance with Method 3001, test condition I of MIL-STD-1344. The following details and exceptions shall apply:

a. Test voltage: 350 volts alternating current (VAC) 60 hertz (Hz), or 500 VDC
b. Test points: Between adjacent contacts on double-sided boards

4.7.3 Mechanical testing.

4.7.3.1 Engaging and separation force (see 3.5.5). Compliant pins shall be tested in accordance with Method 2014 of MIL-STD-1344. The engaging and separation forces shall be determined using the test blade specified in FIGURE 6. The maximum engaging force shall be 10 ounces (oz) the minimum separation force shall be 2 oz.

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GRID SPACING	A ±.0002	B +.002 -.003	C ±.005	D ±.010		F MAX	G ±3
.100	.0200	.050	.164	.040		.008	15

INCHES	MM	INCHES	MM	INCHES	MM
.0005	.01	.0210	.53	.100	2.54
.002	.05	.0300	.76	.125	3.18
.003	.08	.0320	.81	.150	3.81
.005	.13	.040	1.02	.164	4.17
.008	.20	.050	1.27	.200	5.08
.010	.25	.060	1.52	.290	7.37
.0190	.48	.070	1.78	.310	7.87
.020	.51	.083	2.11		

NOTES:

1. MATERIAL: HARDENED TOOL STEEL.
2. HARDNESS: ROCKWELL "C" 50-55.
3. SURFACE FINISH: 4-8 MICRO-INCH ON WORKING SURFACES IAW MIL-STD-10.
4. GRINDING TO BE IN THE DIRECTION OF BLADE MOVEMENT DURING TEST.

FIGURE 6. Engaging and separation force test blade.

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4.7.3.2 Compliant component retention (see 3.5.2). Compliant components shall be tested in accordance with Method 2007 of MIL-STD-1344. After initial insertion, a minimum of 7.5 pounds (33.36 Newton) and a maximum of 45 pounds (200.17 Newton) of vertical force shall be applied to the compliant component tail perpendicular to the printed wiring backplane assembly. The compliant component shall have been displaced from the original position within the range of force specified.

4.7.3.2.1 Conditioning. The compliant component mounting hole in the backplane shall be conditioned by using those pins of the same design supplied by the manufacturer for qualification. The hole shall be conditioned by the insertion and removal of a virgin compliant pin followed by the insertion and removal of a second virgin compliant pin in the same hole. A third virgin compliant pin shall then be inserted in the hole, and the pushout force shall conform to the requirements of 3.5.2.2.

4.7.3.3 Compliant components torque (see 3.5.6). Compliant components shall be tested in accordance with Method 211 of MIL-STD-202, test condition E. Three inch-ounces (0.02118 Newton-meter) of torque shall be applied to the compliant component tail.

4.7.3.4 Housing retention force (see 3.5.1). The capability of the housing to be retained by the compliant pins shall be measured in a test set-up similar to that shown in FIGURE 7. The load required to remove the housing shall be as specified on the applicable specification sheet.

4.7.3.5 Durability testing (see 3.5.11). Compliant components shall be tested in accordance with Method 2016 of MIL-STD-1344. Compliant pins shall be subjected to 500 cycles of mating and unmating.

4.7.3.6 Plated-through hole integrity. After hole conditioning as specified in 4.7.3.2.1, compliant components contained in plated-through holes shall be microsectioned as specified herein to determine conformance to 3.5.3:

a. Double-sided printed wiring boards. Plated-through holes containing compliant components shall be microsectioned in the horizontal plane and examined for conformance to 3.5.3. At least two different levels of hole depth shall be viewed and measurements shall be taken at the entrance to the hole and half-way down the compliant section. It is not mandatory that the same hole be viewed at both levels.
b. Multilayer printed wiring boards. Plated-through holes containing compliant components shall be microsectioned in both the vertical and horizontal planes. In both the horizontal and vertical planes the microsectioned sample shall be examined for conformance to 3.5.3. On the vertical plane, the sample shall additionally be viewed to ensure that no copper cracks, separations between conductive interfaces, or laminate-to-copper separations have occurred.

Samples shall be viewed at magnification of 400X to resolve questionable copper thickness interfaces between the pin and printed wiring laminate.

4.7.4 Environments.

4.7.4.1 Thermal shock (see 3.5.7). Connector component parts mounted to the qualification test sample shall be tested in accordance with Method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition: B.
b. Following testing, connectors shall conform to the requirements of 3.5.4.2.

4.7.4.2 Moisture resistance (see 3.5.12). Connector component parts mounted to the qualification test sample shall be tested in accordance with Method 106 of MIL-STD-202. The following exception shall apply:

a. Steps 7A and 7B are not required.

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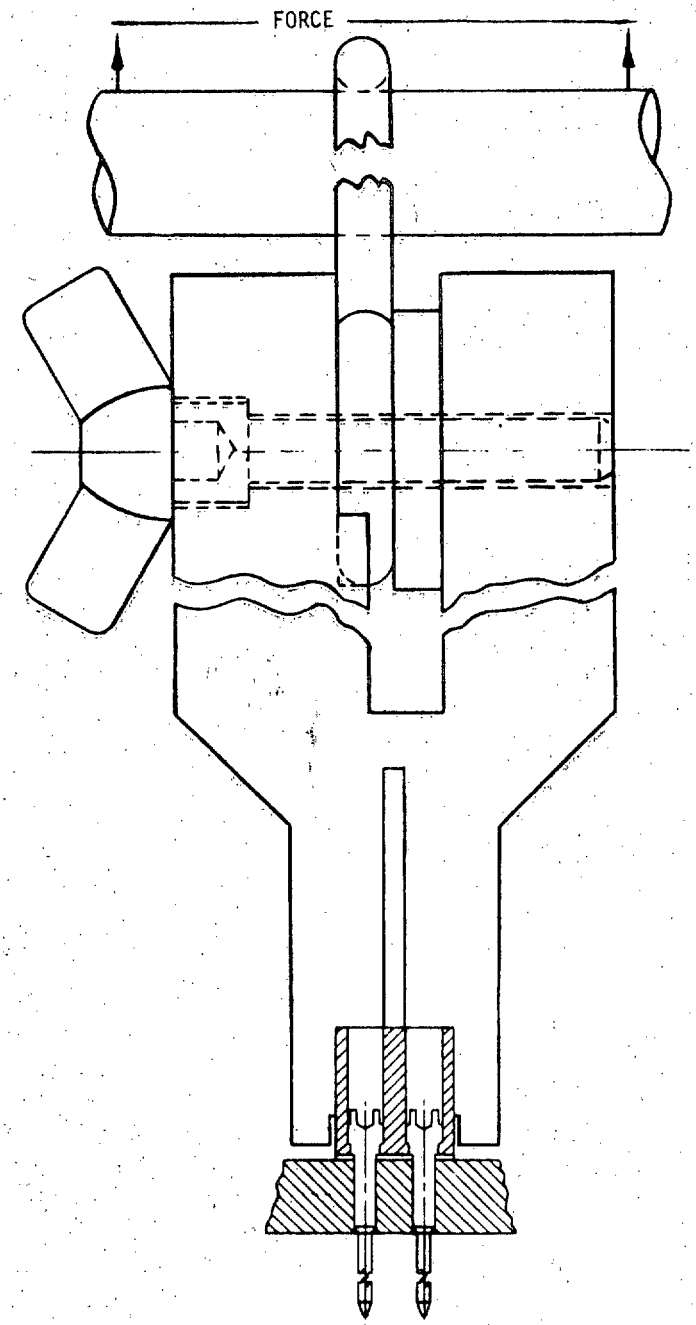


FIGURE 7. Test set-up for housing retention force.

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4.7.4.3 Temperature-altitude testing (see 3.5.14). Connector component parts mounted to the printed circuit board shall be tested in accordance with MIL-STD-810, Method 504.1, Category 6, Steps 1B, 3, 6, and 13. The following details and exceptions shall apply:

a. Step 3: 30 minutes. Test potential of ± 30 VDC applied between all adjacent contacts.
b. Step 6: Last half-hour of test shall have a test potential of ± 30 VDC applied between all adjacent contacts.
c. Step 13 : Temperature shall be 60°C.

4.7.4.4 Vibration testing (see 3.5.8). Connector component parts mounted to the qualification test sample with an appropriate connector and supporting structure shall be tested in accordance with Methods 204 and 214 of MIL-STD-202. The following details and exceptions shall apply:

a. Discontinuity shall be monitored during testing (1.0 μs maximum).
b. Sinusoidal test condition: G
c. Duration of sinusoidal test: Three cycles each axis.
d. Random test condition: I (test condition letter E).
e. Duration of random test: 30 minutes each axis.

4.7.4.5 Shock testing (see 3.5.9). Connector component parts mounted to the qualification test sample with an appropriate connector and supporting structure shall be tested in accordance with Method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Discontinuity shall be monitored during testing (1.0 μs maximum)
b. Test condition: I

4.7.4.6 Salt spray testing (see 3.5.10). Connector component parts mounted to the qualification test sample shall be tested in accordance with Method 101 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition letter: B
b. Conditioning after exposure: All exposed surfaces of the connector component parts shall be thoroughly washed with warm tap water and air dried.
c. Subsequent to exposure: A visual inspection shall be conducted.

4.7.4.7 Life (see 3.5.15). Connector component parts mounted to the qualification test sample shall be tested in accordance with Method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Test condition: +125 ±3°C.
b. Test time condition: D

4.7.4.8 Gas-tight-joint. Compliant pin to printed wiring board interface gas-tight area shall be tested as follows:

a. The samples shall be suspended and sealed in a container with an appropriate volume of aqua regia solution (1:1 concentrated hydrochloric and nitric acids). The solution shall not touch the assemblies. The assemblies shall be exposed to the aqua regia fumes for 30 minutes.
b. Dry the samples and test in accordance with 4.7.2.2

5. PACKAGING

5.1 Preparation for delivery shall be in accordance with MIL-C-55330.

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6. NOTES

6.1 Intended use. Connector component parts covered by this specification are intended for use in printed wiring backplanes applicable to ground support, airborne and shipboard electrical and electronic equipment.

6.2 Current-carrying-capacity. The current ratings given in the specification sheets are for single contacts. It may be necessary to derate the total current carrying capacity of a connector.

6.3 Contact insertion and removal tool. When applicable, a suitable hand tool or tools will be provided with qualification samples to insert and remove the contacts(see FIGURES 8 and 9).

6.4 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Title, number, and date of the applicable specification sheet and the complete part number
c. Tools (as required)
   1. Insertion and withdrawal tools
d. Marking, levels of preservation and packaging and packing required (see 5.1)

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Electronic Systems Command, ELEX 5043, Department of the Navy, Washington, D.C. 20360. Information pertaining to qualification of products may be obtained from either the Naval Electronic Systems Command or the Defense Electronics Supply Center (DESC), Dayton, OH 45444, agent for administration of the Qualified Products List. Application for qualification tests shall be made in accordance with Provisions Governing Qualification SD-6 (see 6.5.1). The printed wiring backplane assemblies, compliant pins, and housings shall be forwarded to the Naval Avionics Center, 6000 East 21st Street, Indianapolis, IN 46218, Code 442.

6.5.1 Copies of Provisions Governing Qualification may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.6 Shipment of contacts. At the option of the contracting officer compliant pins may be mounted on a carrier, or delivered loose in bulk.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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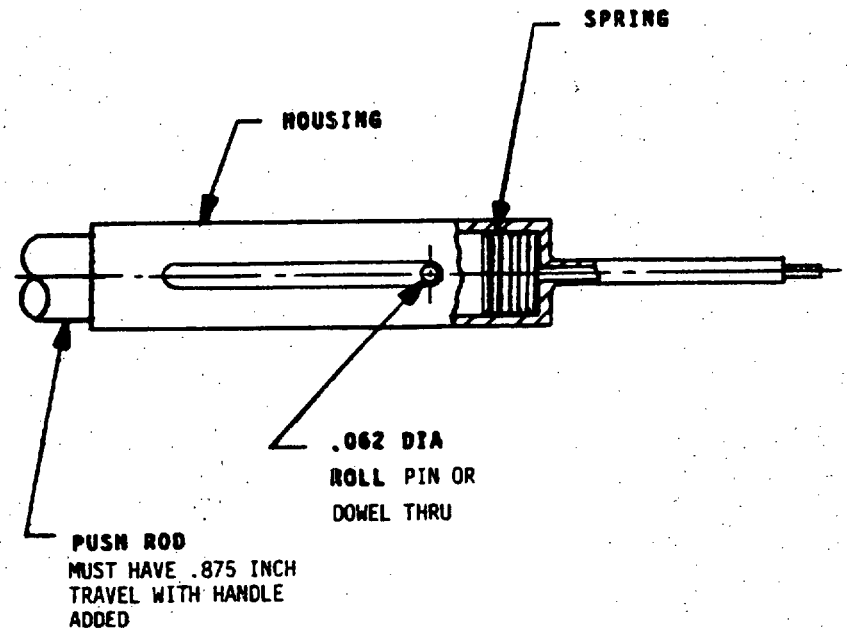
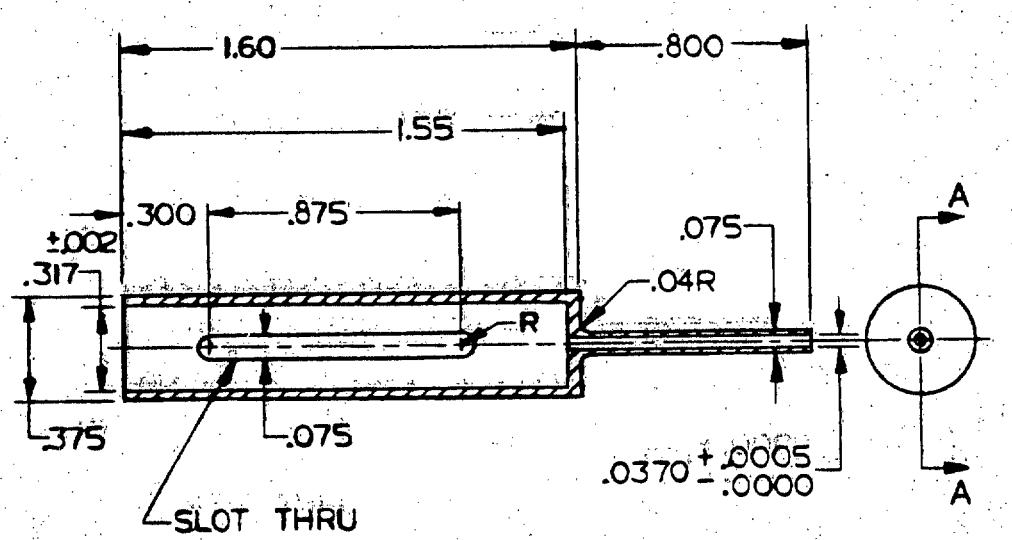


FIGURE 8. Hand removal tool.

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HOUSING



SECTION A-A

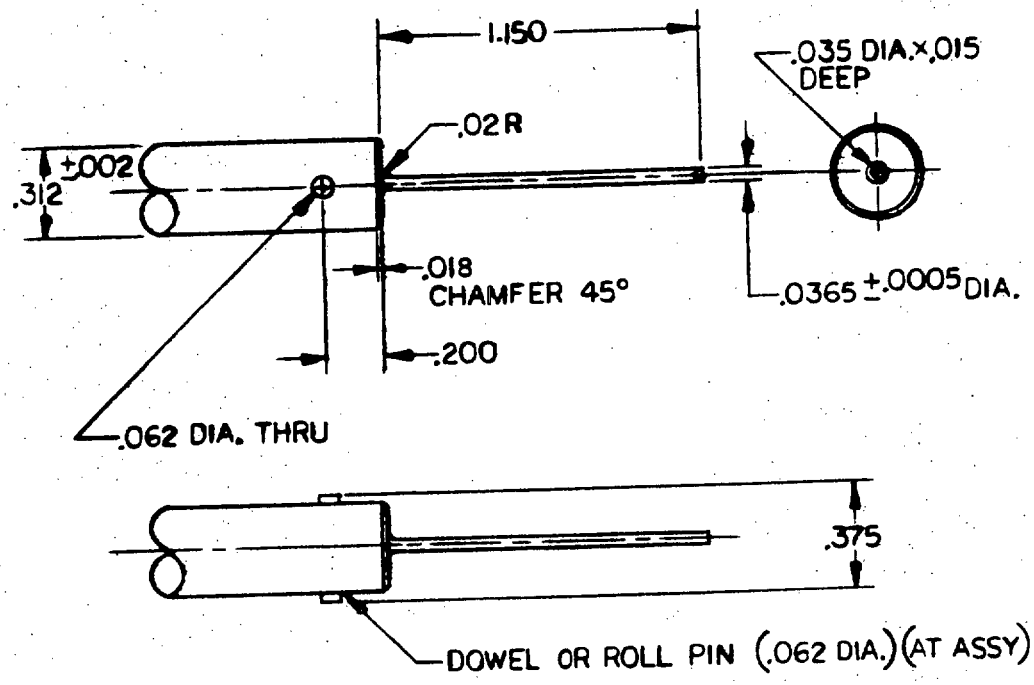
N O T E S :

1. TOL ON .000, ± .005; ON .00, ± .01 UNLESS OTHERWISE SPECIFIED.
2. MATL TO BE: TOOL STEEL ALLOY, QQ-T-570, TYPE D2.
3. HARDEN TO ROCKWELL "C" 50 MIN. AFTER MACHINING.
4. PASSIVATION TREATMENT, QQ-P-35.

FIGURE 8. Hand removal tool. (Continued)

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PUSH ROD



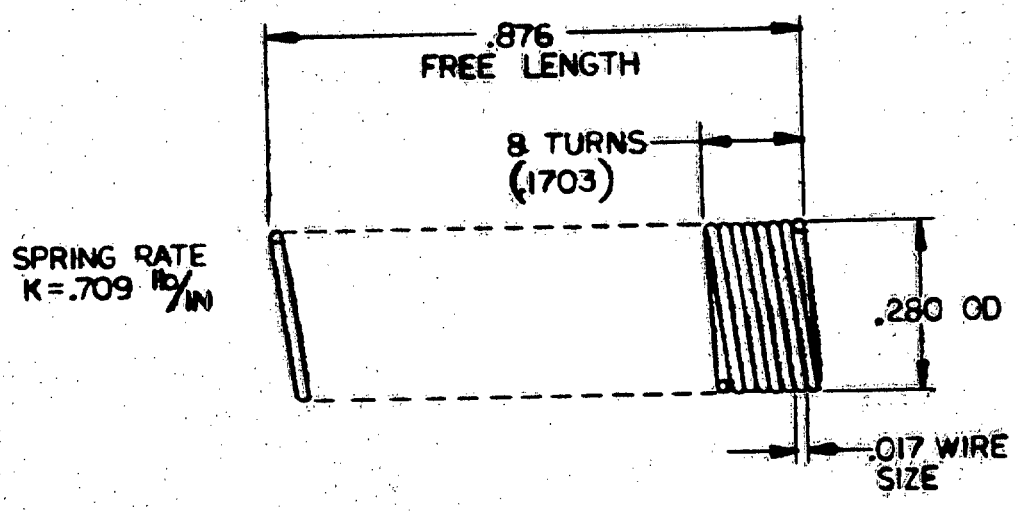
NOTES:

1. TOL: ON .000 ± .005, ON .00 ± .01.
2. MATL TO BE: TOOL STEEL ALLOY, QQ-T-570, TYPE D2 HARDEN TO ROCKWELL "C" 50 MIN. AFTER MACHING.
3. FINISH: DRY FILM LUBRICATE SLIDING SURFACES.
4. ADD HANDLE FOR EASE OF USE. COMPLETED OVERALL LENGTH 5-6 INCHES.

FIGURE 8. Hand removal tool. (Continued)

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SPRING

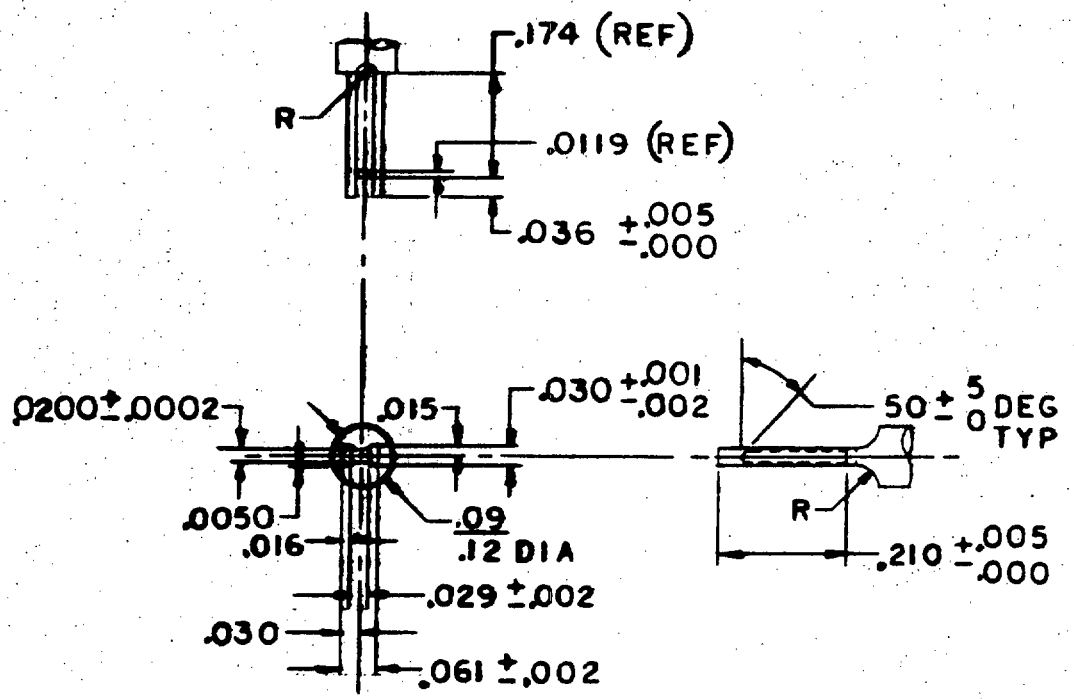


N O T E S :

1. MATL TO BE: STAINLESS STEEL AISI 301 OR 302.
2. HEAT TREAT AS REQD TO OBTAIN SPRING RATE.
3. DEGREASE.

FIGURE 8. Hand removal tool. (Continued)

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NOTES:

1. MATL: TOOL STEEL ALLOY, QQ-T-570, TYPE D2
2. HARDEN TO ROCKWELL "C" 50 MIN. AFTER MACH'N'G.
3. ADD HANDLE FOR EASE OF USE. COMPLETED OVERALL LENGTH 5-6 INCHES.
4. BREAK ALL SHARP CORNERS & EDGES.

FIGURE 9. Hand insertion tool.

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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